University of Glasgow

April/May Exam Session 2024
Duration: 1 hour 30 minutes
Additional time: 30 minutes
Timed exam fixed start time

DEGREES OF MSc, MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

TEXT AS DATA FOR MSC/MSCI
COMPSCI 5096/5106

(Answer all 3 questions)

This examination paper is an open book, online assessment and is worth a total of 60 marks.

1. Question on Tokenisation. (Total marks: 20)

(a) Why are stopwords normally removed when encoding text in a bag-of-words representation? Why are stopwords normally **not** removed when encoding text with a transformer network? Provide an illustrative example.

[4]

(b) One of your colleagues states that the `[UNK]` token is not necessary for subword tokenisers like BPE (Byte Pair Encoding). Considering what you know about Unicode encoding, are they correct? Why or why not?

[4]

(c) Perform sub-word tokenisation over the following two tokens using the following BPE rules. Assume all characters `[m,i,s,p,e]` exist in the vocabulary. Show all your steps.

[6]

BPE Rules:

```
i, s -> is
p, i -> pi
i, e -> ie
p, ie -> pie
p, pi -> ppi
si, ppi -> sippi
```

Tokens to encode:

```
mississippi pie
```

(d) What would be the effect of setting BPE's target vocabulary size too low (e.g., 100) when training? What about too high (e.g., 100,000,000)?

[6]

2. Question on Language Modelling with Character N-grams. (Total marks: 20)

Consider the text below that is 20 characters long. We will build n-gram language models using this text and then generate new text to follow it.

BCBBCBBABCCCBBCCBCCB

(a) Generate the next three tokens for the 20 character text above. Use a character **unigram** model and a **greedy approach**

[2]

(b) Generate the next three tokens for the 20 character text above. Use a character **trigram** model and a **greedy approach**. You do not need to calculate the full trigram model.

[6]

(c) Generate the next **two** tokens for the 20 character text above. Use a character **bigram** model and **beam search** with a beam width of 2. Show the calculations and active beams at each step and then the final output with the highest probability.

[8]

(d) Propose a two character text that a bigram model would be unable to generate (following the 20 character text above). It should use the same alphabet as before. Explain why a model is unable to generate it. Briefly propose a method for adjusting the probabilities and the text generation method which means that the language model may generate it.

[4]

3. Question on Indirect Answer Classification. (Total marks: 20)

Recall the task of Indirect Answer Classification (IAC) from the paper *"Id rather just go to bed": Understanding Indirect Answers* (Louis et al., EMNLP 2020), which you read for Coursework 2. (The paper is available on Moodle if you need to refer back to it.)

(a) Consider a situation where you try a simple prompt on a large language model to perform IAC on *Circa*, and you obtain a higher effectiveness on the test set than what was reported by Louis et al. What do you need to consider before asserting that the effectiveness is "zero-shot" (obtained without training the model without any IAC data)?

[2]

(b) A colleague suggests that *Circa*'s Fleiss kappa of 0.61 means that the dataset's labels are unreliable. They suggest using a large language model to provide the labels instead. Provide three counter-arguments.

[6]

(c) Recall that the *unmatched* evaluation setting aims to evaluate a model's performance on unseen scenarios. Using what you learned about text similarity, why might their approach be insufficient? Propose a technique that could better evaluate performance on unseen scenarios.

[6]

(d) The paper tests models that use multiple stages of fine-tuning on different datasets. What is the motivation for this approach? Why is effectiveness degraded when fine-tuning on some datasets?

[6]

END OF QUESTION PAPER